

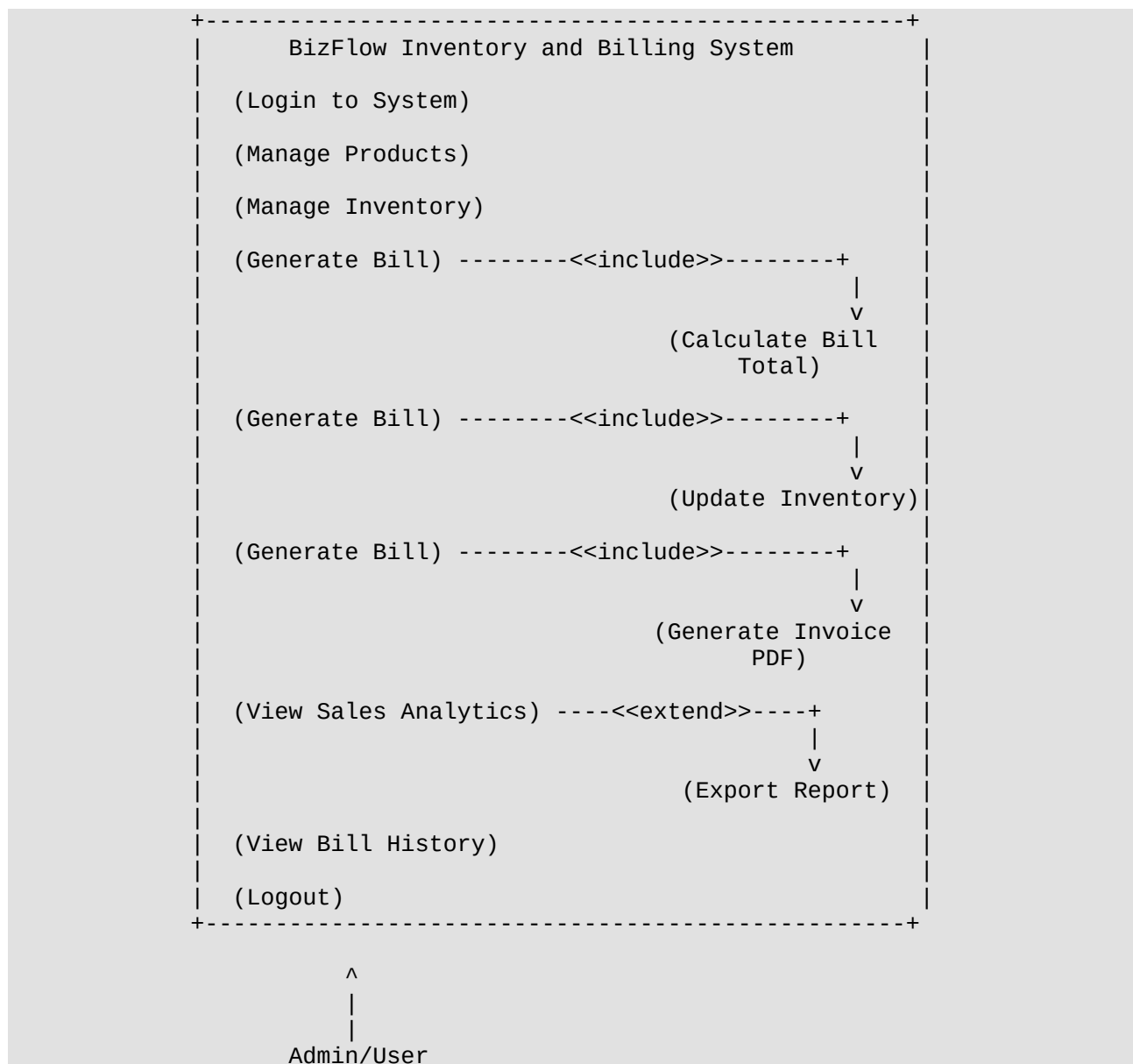
Experiment :- 3

AIM :- Explain any one Use Case with its Situation and Diagram.

A Use Case describes how a user (actor) interacts with a system to achieve a specific goal.

It helps in understanding:

- Functional requirements
- User interaction
- System behavior



Suggested Diagram Components:

Actor:

Admin/User

Use Cases:

- Login to System
- Manage Products
- Manage Inventory
- Generate Bill
- Update Inventory
- Generate Invoice PDF
- View Sales Analytics
- View Bill History
- Logout

Relationships:

Generate Bill <> Calculate Bill Total

Generate Bill <> Update Inventory

Generate Bill <> Generate Invoice PDF

View Sales Analytics <> Export Report

In the BizFlow Inventory and Billing Management System, the main purpose is to manage products, track inventory, generate bills, and maintain sales records efficiently.

Explanation of Diagram :-

The Use Case Diagram shows the interaction between:

- Admin/User

- Product Management Module
- Inventory Management Module
- Billing System
- Invoice Generation Module
- Analytics Dashboard

Main Functions Shown:

1. Login to System
2. Manage Products
3. Manage Inventory
4. Generate Bill
5. Generate Invoice PDF
6. View Sales Analytics
7. View Bill History
8. Logout

BizFlow Inventory and Billing Management System:

i) Generate Bill and Update Inventory

Description:-

This use case explains how the system allows the user to create a sales bill, calculate the total amount, update inventory automatically, and generate an invoice.

The Admin/User performs billing operations using the BizFlow dashboard.

Actors Involved :-

i) Primary Actor

Admin/User

ii) Secondary Actors

- Product Catalog Module
- Inventory Management Module
- Invoice Generation Module

Objectives of the Use Case :-

The objectives are:

- Generate customer bills
- Calculate bill totals
- Update inventory automatically
- Generate invoice PDFs
- Store bill records
- Track sales activities

Main Flow of Events (Step-by-Step Working) :-

Step 1: User Login

The Admin/User logs into the BizFlow system dashboard.

Step 2: Select Products

The user selects products from the product catalog available in the system.

Step 3: Enter Quantity

The quantity of each selected product is entered for billing.

Step 4: Bill Calculation

The system calculates:

- Product-wise amount
- Total quantity
- Total bill amount
- Applicable taxes

Step 5: Inventory Update

After bill confirmation, the inventory quantity is automatically reduced according to the products sold.

Step 6: Bill Storage

The generated bill is stored securely in the database for future reference.

Step 7: Invoice Generation

The system generates a professional PDF invoice containing:

- Bill Number

- Product Details
- Quantity
- Rate
- Total Amount
- Date and Time

Step 8: Analytics Update

Sales data is updated automatically in the analytics dashboard.

Step 9: View Bill History

The user can view previously generated bills and invoices whenever required.

Step 10: Logout

The user logs out from the system after completing operations.

Benefits of the Use Case :-

i) Faster Billing Process

Bills can be generated quickly and accurately.

ii) Automatic Inventory Management

Stock levels are updated without manual intervention.

iii) Accurate Record Keeping

All bills and sales transactions are stored securely.

iv) Better Business Monitoring

Sales analytics help in tracking business performance.

v) Reduced Human Errors

Automation minimizes mistakes in calculations and inventory tracking.